



Online Shopping Behavior Analysis

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Introduction to Statistical Analysis

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Abstract

This project presents a complete statistical analysis of an Online Shopping Behavior dataset using Python. The main objective of the project was to understand customer purchasing behavior and identify relationships between browsing activity, customer demographics, and purchase patterns through a structured data analysis pipeline.

The project was implemented in several stages, beginning with data understanding and preprocessing. During the cleaning phase, missing values, inconsistent text entries, duplicate records, and logical errors such as negative numerical values were identified and corrected to ensure data quality and analytical reliability.

After preprocessing, descriptive statistical analysis was performed to summarize the dataset using measures such as mean, median, standard deviation, minimum, and maximum values. Correlation analysis was also conducted to investigate relationships between important variables including age, pages viewed, time spent on the website, and purchase amount.

Furthermore, multiple visualization techniques such as histograms, scatter plots, box plots, pie charts, and correlation heatmaps were created to transform numerical findings into clear visual insights. The results showed that customer purchase behavior is influenced by multiple factors rather than a single variable, while mobile devices represented the most commonly used shopping platform.

Finally, the project provided several business recommendations related to improving user experience, customer engagement, and retention strategies. This project demonstrates the importance of data preprocessing, statistical analysis, and visualization in extracting meaningful insights from real-world datasets.

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1. Introduction

Understanding customer behavior is one of the most important aspects of modern e-commerce systems. Online shopping platforms generate large amounts of customer interaction data every day, including browsing activity, purchasing patterns, and user engagement.

The purpose of this project is to perform a complete statistical analysis on an Online Shopping Behavior dataset using Python. The project follows the main stages of the data analysis pipeline including:

- Data Understanding
- Data Cleaning
- Statistical Analysis
- Data Visualization
- Business Insights and Recommendations

Through this project, we aim to explore customer behavior and identify relationships between different variables affecting online purchasing decisions.

2. Dataset Description

The dataset contains information related to customer behavior on an online shopping platform

Dataset Variables

Variable	Description
age	Customer age
gender	Customer gender
time_on_site	Time spent on the website
pages_viewed	Number of viewed pages
device	Device type
purchase_amount	Purchase value
discount_used	Whether a discount was used
click_rate	Customer click rate
return_customer	Whether the customer is returning or new

3. Team Roles & Responsibilities

3.1 Besan Ahmad – Data Preprocessing & Documentation Lead

As the team member responsible for data quality and project organization, my role focused on preparing the dataset for statistical analysis and organizing the final project report and presentation.

Responsibilities

Data Auditing

- Inspected the raw dataset.
- Identified missing values, duplicates, and inconsistent data entries.
- Examined dataset structure and data types using Pandas.

Handling Missing Values

- Applied statistical imputation methods:
 - Mean for numerical columns.
 - Mode for categorical columns.
- Ensured data completeness before analysis.

Text Standardization

- Standardized inconsistent text entries.
- Converted text values to lowercase.
- Replaced invalid placeholders such as:
 - ?
 - unknown
 - blank spaces

Logical Error Correction

- Corrected negative values found in:
 - time_on_site
 - click_rate
- Applied absolute value transformation to maintain logical consistency.

Data Encoding & Formatting

- Converted binary categorical values into numerical format (0/1).
- Unified data types for analysis compatibility.

Final Validation & Export

- Removed duplicate rows using drop_duplicates().
- Performed final validation checks for:
 - Missing values
 - Data consistency
 - Correct data types
- Exported the final cleaned dataset as clean_data.csv.

Report & Presentation Preparation

- Organized and formatted the final project report.
- Prepared the project presentation slides.
- Ensured consistency between all project phases.

3.2 Jana Shahruri – Statistical Analysis Lead

As the statistical analyst, my role focused on performing descriptive and correlation analysis to extract meaningful insights from the cleaned dataset.

Responsibilities

Descriptive Statistics

- Applied descriptive statistical measures including:
 - Mean
 - Median
 - Standard deviation
 - Minimum and maximum values
 - Quartiles
- Used `df.describe()` to generate statistical summaries.

Spending Behavior Analysis

- Analyzed the `purchase_amount` variable in detail.
- Compared mean and median values to evaluate data balance.
- Studied spending variability using standard deviation.

Correlation Analysis

- Calculated correlation coefficients between key numerical variables:
 - Age vs purchase amount
 - Pages viewed vs purchase amount
 - Time on site vs purchase amount
- Built the complete correlation matrix using `.corr()`.

Statistical Interpretation

- Interpreted relationships between variables.
- Concluded that customer purchase behavior depends on multiple combined factors.

Tools Used

- Python
- Pandas
- NumPy

- Descriptive Statistics
- Correlation Analysis

3.3 Hala Al-Masri – Data Visualization & Business Insights Lead

As the visualization and business insights specialist, my role focused on transforming statistical findings into visual representations and actionable business recommendations.

Responsibilities

Data Visualization

- Designed multiple visualizations including:
 - Histograms
 - Pie Charts
 - Scatter Plots
 - Box Plots
 - Correlation Heatmaps

Visual Relationship Analysis

- Visualized relationships between:
 - Gender and purchase amount
 - Device type and customer behavior
 - Time on site and purchase amount

Correlation Heatmap

- Created a visual heatmap for the correlation matrix to simplify statistical interpretation.

Business Recommendations

- Converted analytical findings into practical business recommendations related to:

- Mobile user experience
- Customer retention
- Discount strategies
- Customer engagement improvement

Final Interpretation

- Connected statistical findings with real-world e-commerce decisions.

4. Methodology

The project followed a structured data analysis workflow consisting of:

1. Data Understanding
2. Data Cleaning & Preprocessing
3. Statistical Analysis
4. Data Visualization
5. Business Insights & Recommendations

The following Python libraries were used:

- Pandas
- NumPy
- Matplotlib

5. Data Cleaning & Preprocessing

Data cleaning was one of the most important stages of the project because raw datasets often contain inconsistencies and errors.

5.1 Handling Missing Values

Missing values were handled using statistical imputation methods:

- Numerical columns → Mean
- Categorical columns → Mode

This ensured dataset completeness without losing records.

5.2 Replacing Invalid Values

Invalid placeholders such as:

- ?
- unknown
- blank spaces

were replaced with NaN before preprocessing.

5.3 Text Standardization

Text values were standardized into lowercase format to prevent duplicated categories.

Examples:

- Male → male
- Desktop → desktop

5.4 Logical Error Correction

Negative values were found in:

- time_on_site
- click_rate

Since these values cannot logically be negative, absolute values were applied.

5.5 Removing Duplicates

Duplicate rows were removed to improve data quality and avoid biased results.

6. Statistical Analysis

A descriptive statistical analysis was performed to better understand customer behavior.

The analysis included:

- Mean
- Median
- Standard deviation
- Minimum and maximum values
- Quartiles

Special focus was placed on the purchase_amount variable because it directly represents customer spending behavior.

The results showed that the mean and median were relatively close, indicating balanced spending behavior and limited influence of extreme outliers.

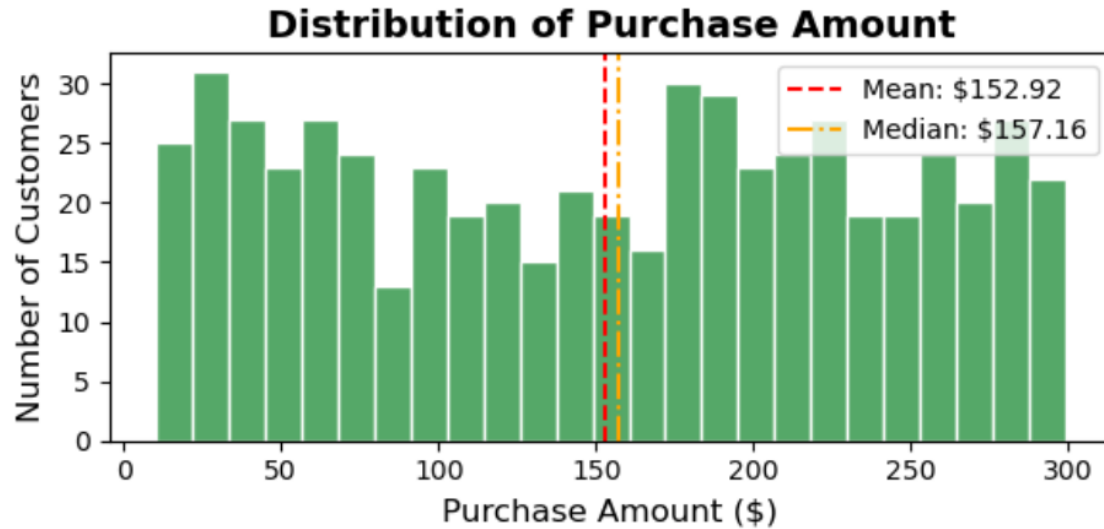


Figure 1: Distribution of Purchase Amount

This histogram shows the distribution of customer purchase amounts. It indicates that most customers have moderate spending behavior, with values distributed across a wide range. The mean (\$152.92) and median (\$157.16) are very close, which suggests that the data is relatively balanced and not heavily skewed by extreme values.

7. Correlation Analysis

Correlation analysis was conducted to study relationships between numerical variables.

The following relationships were analyzed:

- Age vs purchase amount
- Pages viewed vs purchase amount
- Time on site vs purchase amount

The following heatmap illustrates the correlation between all numerical variables in the dataset.

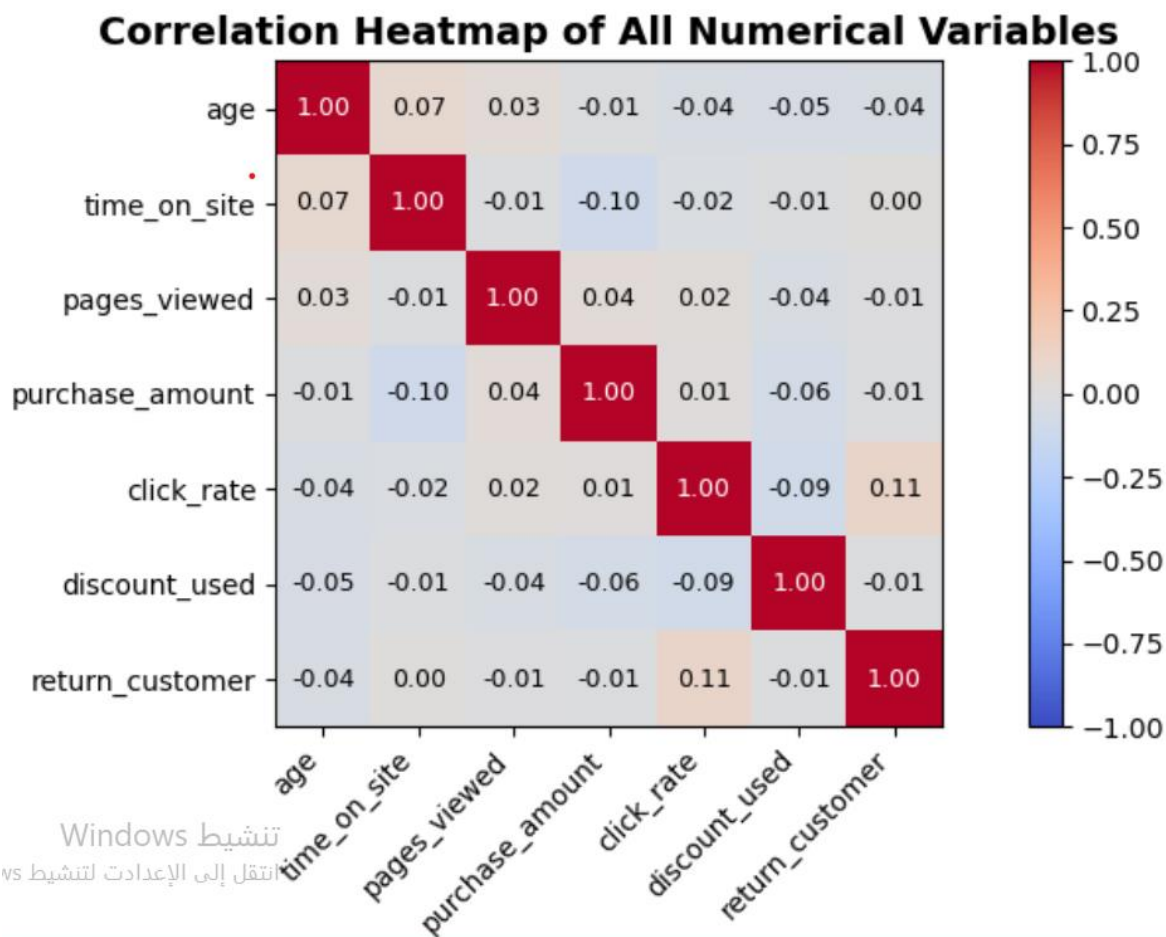


Figure 7: Correlation Heatmap of Numerical Variables

This heatmap summarizes the correlation coefficients between key variables including age, time on site, pages viewed, purchase amount, click rate, discount usage, and customer type. The results show that most correlations are weak, with values close to zero. For example, time on site and purchase amount have a correlation of -0.10, while age and purchase amount show -0.01. This indicates that customer purchasing behavior is influenced by multiple combined factors rather than a single variable.

The correlation matrix indicated that most relationships were weak, suggesting that customer purchasing behavior depends on multiple combined factors rather than a single

variable. The heatmap confirms that no single variable strongly predicts purchase behavior.

8. Data Visualization

Several visualization techniques were used to simplify interpretation and present findings clearly.

The visualizations included:

- Histograms
- Scatter plots
- Pie charts
- Box plots
- Correlation heatmaps

These visualizations transformed numerical findings into understandable visual insights

The following chart compares customer spending behavior across genders.

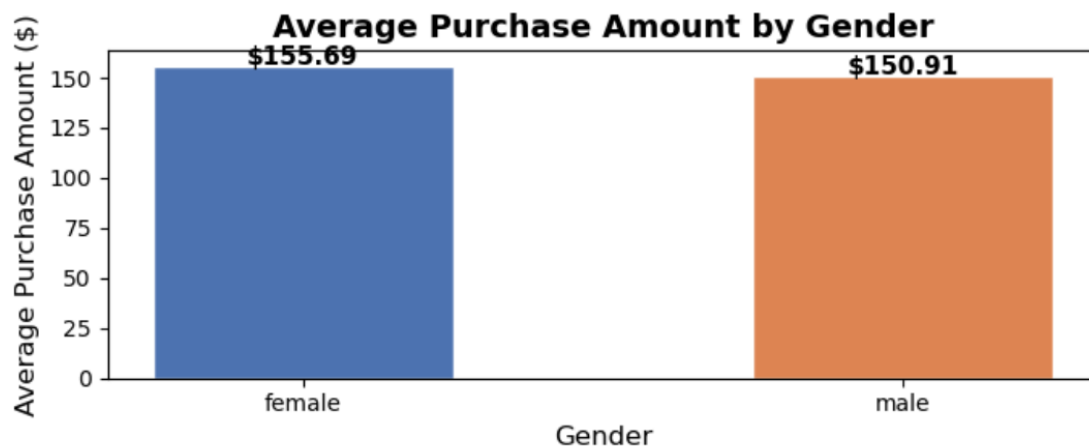


Figure 2: Average Purchase Amount by Gender

This bar chart compares the average purchase amount between female and male customers. The results show that female customers have a slightly higher average purchase amount (\$155.69) compared to male customers (\$150.91). However, the difference is small, indicating that gender does not have a strong influence on purchasing behavior in this dataset

The following chart illustrates the preferred device used by customers during online shopping.

Device Usage Distribution

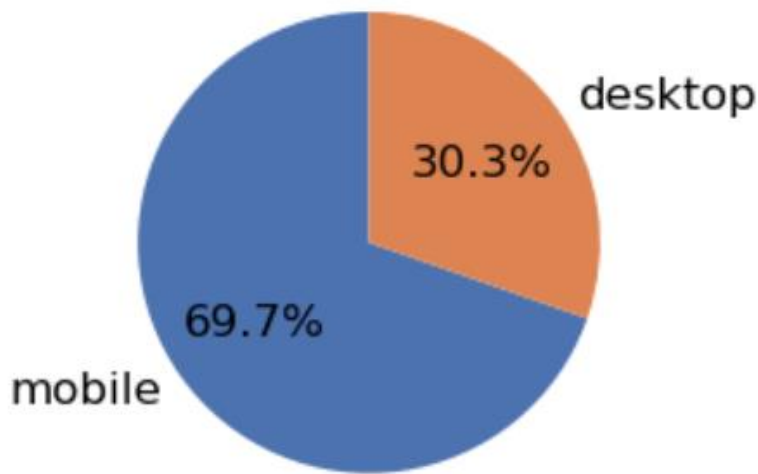


Figure 3: Device Usage Distribution

This pie chart shows the distribution of devices used by customers for online shopping. The results indicate that mobile devices dominate usage with 69.7%, while desktop users represent 30.3%. This clearly shows that most customers prefer mobile platforms, highlighting the importance of optimizing the shopping experience for mobile users

The following scatter plot illustrates the relationship between time spent on the site and purchase amount.

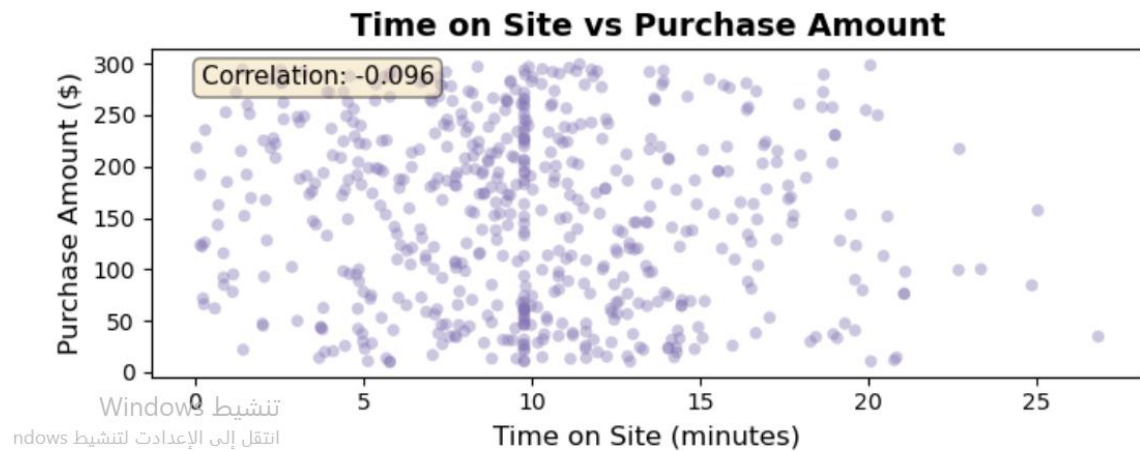


Figure 4: Time on Site vs Purchase Amount

This scatter plot shows the relationship between time spent on the website and the purchase amount. The correlation coefficient (-0.096) indicates a very weak negative relationship, meaning that the amount of time a customer spends on the site does not strongly influence how much they spend. This suggests that other factors, such as customer demographics or device type, play a more important role in purchasing behavior.

The weak correlation confirms that time on site is not a strong predictor of spending behavior.

The following box plot illustrates the effect of discount usage on customer purchase amounts.

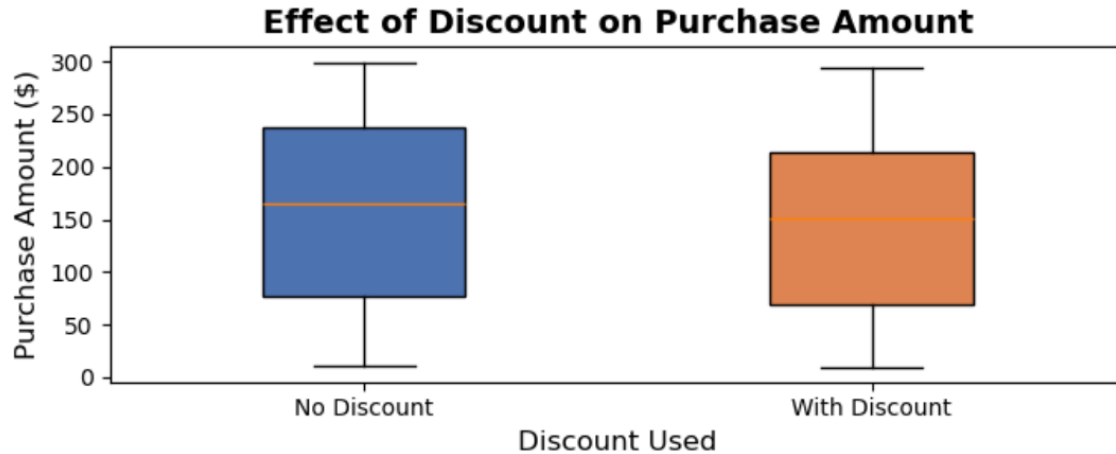


Figure 5: Effect of Discount on Purchase Amount

This box plot compares the distribution of purchase amounts between customers who used a discount and those who did not. The results show that the median purchase amount is slightly higher for customers without discounts compared to those with discounts. This suggests that discounts do not necessarily lead to higher spending, and in some cases, customers without discounts may spend more.

Discounts are not a strong driver of higher spending.

The following chart compares the average purchase amount between new and returning customers.

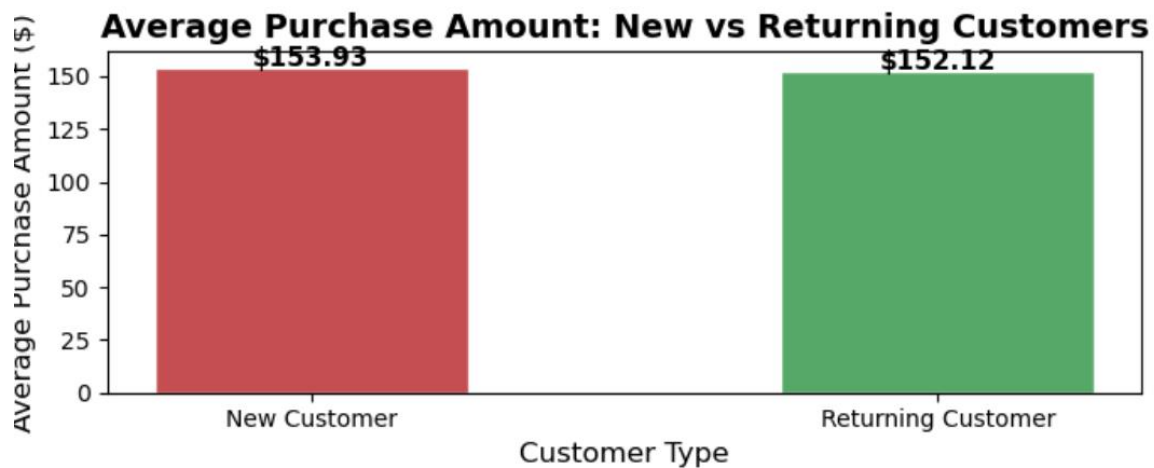


Figure 6: Average Purchase Amount – New vs Returning Customers

This bar chart compares the average purchase amount between new and returning customers. The results show that new customers spend slightly more on average (\$153.93) compared to returning customers (\$152.12). However, the difference is small, indicating that customer type (new vs returning) does not strongly influence purchasing behavior in this dataset.

New customers spend marginally more, but the difference is negligible.

9. Key Findings

The analysis produced several important findings:

1. Most customers demonstrated moderate spending behavior.
2. Mobile devices were the most commonly used shopping platform.
3. Customer age showed weak correlation with purchase amount.
4. Browsing activity did not strongly influence spending behavior.
5. Data cleaning significantly improved analysis reliability.

10. Recommendations

Based on the analysis results, the following recommendations are suggested:

1. Improve mobile user experience.
2. Increase customer interaction within the platform.
3. Develop personalized recommendation systems.
4. Enhance customer loyalty programs.

11. Challenges

Several challenges were encountered during the project:

- Missing values
- Invalid entries
- Data inconsistency
- Negative numerical values
- Duplicate records

All challenges were successfully handled during preprocessing.

12. Conclusion

This project successfully implemented a complete statistical data analysis pipeline using Python.

The dataset was cleaned, analyzed, and visualized to extract meaningful insights about online customer behavior.

The project highlighted the importance of:

- Data preprocessing
- Statistical analysis
- Data visualization
- Business interpretation

Overall, this project provided valuable practical experience in statistical data analysis and customer behavior understanding.

13. References

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